

- 5 Long-distance transport in mammals and plants requires a transport system and a transport medium.
- (a) The double circulatory system of mammals, which is summarised in Figure 5.1, includes the heart and blood vessels.

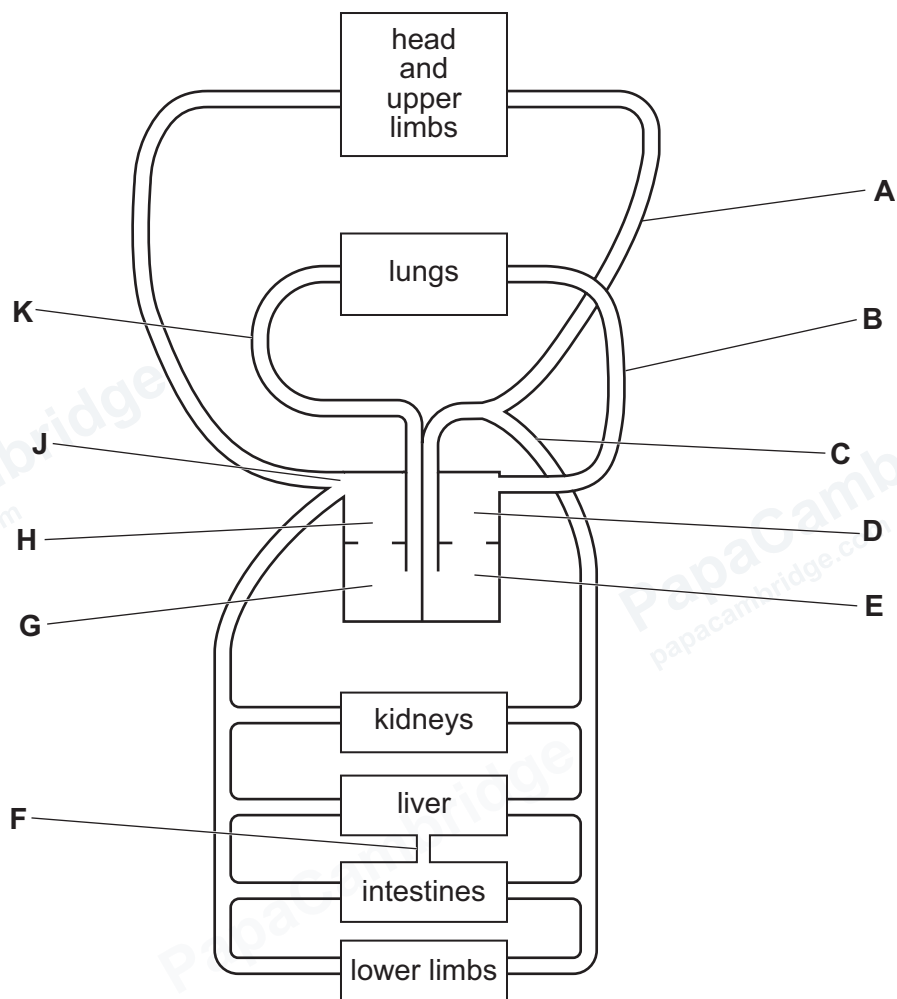


Figure 5.1

Complete Table 5.1 by stating a correct letter from Figure 5.1 that matches each of the descriptions listed.

Table 5.1

description	letter
the chamber of the heart that receives blood from the vena cava	
the chamber of the heart that pumps blood directly into the pulmonary artery	
an artery with a large quantity of elastic fibres and proportionately very little smooth muscle in the tunica media	
a vein transporting oxygenated blood	



(b) Water is the main component of the transport medium in mammals and in plants.

(i) Describe how a high specific heat capacity makes water suitable as the main component of blood.

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(ii) Explain the importance of water in the mass flow of substances in phloem sap.

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[Total: 9]